Ch. 10 — P01 (A) Warmup Analysis

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A simulation analyst runs a single long replication of an M/M/1 queue ($\lambda = 0.7$, $\mu = 1.0$, $\rho = 0.7$) for 10,000 customers starting empty. They divide the run into batches of 100 customers and record the batch-mean wait in queue $\bar{W}_q^{(b)}$ for batches $b = 1, \ldots, 100$.

The theoretical steady-state value is $W_q^* = \lambda/(\mu(\mu - \lambda)) = 7/3 \approx 2.333$ min.

The analyst observes that the first 20 batch means are systematically below W_q^*:

Batch	$\bar{W}_q$
1	0.14
5	0.48
10	1.02
15	1.67
20	2.01
25	2.28
30	2.40

Tasks

1. Explain intuitively why $\bar{W}_q^{(1)} = 0.14$ is so far below $W_q^* = 2.33$ when the system starts empty.
2. Apply Welch's moving average with window $w = 5$ to the table above (compute $\bar{Y}_5(b)$ for batches $b = 5 + 1, \ldots, 25$ using the data given, interpolating the missing batches linearly if necessary). At what batch does $\bar{Y}_5(b)$ first enter the band $[W_q^* \pm 10\%] = [2.10, 2.57]$?
3. The analyst computes the sample mean over:
   (a) All 10,000 customers: $\hat{W}_q = 2.08$
   (b) Customers 2001–10000 (warmup deleted): $\hat{W}_q = 2.31$

   Compute the absolute and percentage bias for each case. Which estimate is unbiased?

4. Suppose the analyst wanted to run independent replications instead of a single long run. With 30 replications of 1,000 customers each and warmup deletion of 200 per replication:

   (a) How many "effective" customers contribute to the estimate across all replications?

   (b) If the standard deviation across replications is $s = 0.42$, compute the 95% CI half-width.

   (c) Does the 95% CI contain $W_q^* = 2.333$?

5. What would happen to the warmup length if ρ were increased from 0.7 to 0.95? Explain qualitatively (no calculation required).